SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ TEST (25 Questions)

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	**Assessment:**	Assessment Tool 3 – MCQ TEST
Weightage:	10% of CIA	**Total Marks:**	25 (25 Questions × 1 Marks)
CO1 Statement:	*Apply appropriate visualization methods to represent distributions, proportions, and relationships in data (BL3).*		
Student Name:	RENEE VINCENT ASR	**Reg. No.:**	192424161
Section / Batch:	BTECH AI & DS	**Date:**	30-0702026

Code of Conduct

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- This test contains 25 MCQ questions, each carrying 1 mark. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Distributions	Analyzing spread, skewness, density, and outliers	1–5	5
x–y Relationships	Exploring relationships and clustering using scatter-based methods	6–10	5
Correlation & Heatmaps	Identifying correlations and density patterns	11–15	5
Temporal Data Representations	Visualizing trends, seasonality, and time-based patterns	16–20	5
Geospatial & Spatio-Temporal Data	Mapping spatial data and combining space with time	21–25	5
		GRAND TOTAL:	**25 Marks**

Annexure A – Multiple Choice Questions (MCQ Test)

1. Which visualization is commonly used to show parts of a whole?
a) Line graph
b) Pie chart
c) Scatter plot
d) Heat map
2. A logistics company maps delivery routes across cities. Why is geospatial visualization preferred?
a) It improves audio quality
b) It represents spatial relationships between locations
c) It removes routing information
d) It reduces geographic accuracy
3. A dataset contains values: 11, 13, 15, 17, 19, 21, 23, 90. Calculate the median and identify the likely skewness.
a) Median = 21, symmetric distribution
b) Median = 18, right-skewed distribution
c) Median = 23, uniform distribution
d) Median = 15, left-skewed distribution
4. A histogram comparing two stores shows one store has a much wider spread of sales values. What analytical conclusion is correct?
a) Spread does not affect performance evaluation
b) Both stores have equal consistency
c) Wider spread implies higher average sales automatically
d) That store experiences greater variability in sales performance
5. A boxplot has the following values: Q1 = 40, Median = 55, Q3 = 70. Calculate the IQR and evaluate the central spread.
a) IQR = 55, indicating uniform spread
b) IQR = 15, indicating no variability
c) IQR = 30, indicating moderate variability in the middle 50%
d) IQR = 70, indicating symmetric distribution
6. A density plot of rainfall data becomes flatter and wider after climate changes over several years. Which analytical interpretation is strongest?
a) Flat density curves indicate absence of observations
b) Rainfall patterns have become more variable and less concentrated
c) Rainfall variability has reduced completely
d) The average rainfall must have decreased
7. A company compares yearly sales distributions. Year 1 → Mean = 500, SD = 40; Year 2 → Mean = 500, SD = 100. Which analytical conclusion is strongest?
a) Mean values alone determine spread
b) Both years have identical distributions
c) Year 1 has greater variability
d) Year 2 sales fluctuate more widely around the average
8. Two distributions are compared using boxplots: Distribution A → Median = 55, Range = 20; Distribution B → Median = 70, Range = 50. Which analytical evaluation is most appropriate?
a) Distribution B has higher central tendency but also greater spread
b) Distribution A has more variability because its range is smaller
c) Both distributions have equal consistency
d) Range cannot indicate spread

9. Two boxplots are compared: Product A → Median lifetime = 8 years, IQR = 1 year; Product B → Median lifetime = 8 years, IQR = 4 years. Which evaluation is most valid?
a) Equal medians imply equal variability
b) IQR does not affect reliability interpretation
c) Product A has more reliable lifetime consistency
d) Product B is more stable because its IQR is larger
10. A dataset contains values: 22, 24, 25, 28, 30, 32, 35, 100. Calculate the median and evaluate the effect of 100 on the distribution.
a) Median = 32, and distribution remains symmetric
b) Median = 25, and 100 creates left skewness
c) Median = 29, and 100 creates right skewness
d) Median = 35, and no skewness exists
11. A histogram for monthly profits shows: 0–10k → 5 months, 10k–20k → 8 months, 20k–30k → 15 months, 30k–40k → 20 months, 40k–50k → 4 months. What analytical conclusion is most suitable?
a) The distribution is strongly left-skewed
b) Profits are concentrated mainly between 30k–40k
c) No concentration pattern exists
d) Profits are uniformly distributed
12. Two violin plots represent employee salaries: Department X → Mean = 50k, SD = 5k; Department Y → Mean = 50k, SD = 20k. Which evaluation is strongest?
a) Department X has more spread because deviation is smaller
b) Violin plots ignore standard deviation
c) Department Y has greater salary variability
d) Both departments have identical distributions
13. A dataset contains the values: 8, 10, 12, 14, 16, 18, 20, 60. Calculate the median and evaluate whether the distribution is skewed.
a) Median = 18, distribution is symmetric
b) Median = 15, distribution is right-skewed
c) Median = 20, distribution is uniform
d) Median = 14, distribution is left-skewed
14. A company visualizes internet traffic between servers using directed graphs. Why are arrows used on edges?
a) To indicate data direction flow
b) To improve brightness
c) To reduce connectivity
d) To hide nodes
15. A researcher compares rainfall patterns over ten years. Which temporal visualization supports trend analysis best?
a) Line chart with time axis
b) Pie chart
c) Word cloud
d) Static matrix
16. Assertion (A): Temporal visualizations can reveal seasonality patterns.
Reason (R): Data values are organized according to time progression.
a) Both A and R are true, and R is the correct explanation of A
b) Both A and R are true, but R is not the correct explanation of A
c) A is true, but R is false
d) A is false, but R is true

17. A navigation application highlights roads with heavy traffic in red. What geospatial principle is applied here?
a) Spatial intensity encoding
b) Random clustering
c) Audio normalization
d) Binary formatting
18. In a network visualization of airline routes, hubs have many incoming and outgoing edges. What does this indicate?
a) Low connectivity
b) High network importance
c) Data corruption
d) Reduced traffic
19. A company compares two delivery systems: System A → Mean delivery time = 25 min, SD = 3; System B → Mean delivery time = 25 min, SD = 10. Which analytical conclusion is most valid?
a) Both systems have identical variability
b) Standard deviation cannot measure consistency
c) System A is more consistent in delivery performance
d) System B is more stable because it has higher deviation
20. A histogram shows frequencies of product defects: 0–2 defects → 60 machines, 2–4 defects → 25 machines, 4–6 defects → 10 machines, 6–8 defects → 5 machines. What analytical conclusion should be drawn?
a) Defects are uniformly distributed
b) Most machines produce high defects
c) The histogram is bimodal
d) Most machines produce very few defects, indicating right-skewness
21. A hospital analyzes patient waiting times using a histogram and notices that most values are concentrated in lower intervals while a few extend far to the right. What analytical interpretation can be made from this distribution?
a) Waiting times follow a right-skewed distribution with a few extreme delays
b) All patients experience identical waiting times
c) The data follows perfect symmetry
d) No variability exists in the dataset
22. A financial analyst uses a violin plot to compare monthly investment returns of two portfolios. One violin plot shows two clear density peaks. What does this most likely indicate about the portfolio returns?
a) The returns contain two dominant groups or patterns
b) The portfolio has zero variability
c) The returns are uniformly distributed
d) The dataset contains only outliers
23. Why do analysts use scatter plots for x–y data?
a) To identify trends and relationships
b) To display percentages only
c) To organize files
d) To remove variables
24. How does visualizing amounts help decision-making?
a) By making quantity comparisons easier
b) By reducing information
c) By hiding differences
d) By avoiding analysis

25. Why are distributions important in data analysis?
a) They help understand how values are spread
b) They remove patterns
c) They reduce accuracy
d) They avoid comparisons

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect		
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers		
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts		
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time		
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________